

Optimizing Small F&B Businesses with Data Analytics

A Proposal report for the BDM capstone Project

Submitted by

Name: Piyush Jha

Roll Number: 21F1002373



IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project Title "*Optimizing Small F&B Businesses with Data Analytics*". I extend my appreciation to "*Shubham Sweet Corner*", for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of the candidate:



Name: Piyush Jha

Date: 11/05/2025

1. Executive Summary and Title

The Project focuses on a small food business located at Lakshminiya Chowk, Polytechnic Bypass Road, Saharsa, Bihar. The business is a B2C and deals in the segment of Food and Beverages.

The major business issues that the organization is facing are related to lack of record-keeping, leading to unclear profits, difficulty in retaining workers, and limited customer reach due to location.

The issues will be addressed by collecting and analyzing the data using simple tracking methods, identifying labor trends, and studying customer patterns to improve business decisions. Simple data visualization techniques will help in understanding sales, expenses, and workforce stability.

The expected outcome helps the organization to track finances, improve worker retention, and attract more customers, leading to better profitability.

2. Organization Background

The organization that I am working with is Shubham Sweet Corner; which is a small for-profit business in food and beverage sector based in Polytechnic Bypass Road, Saharsa, Bihar. The shop was inaugurated in 2020, by Chandra Kishor Jha. The shop specializes in Indian snacks and sweets with some cold refreshments on the side.

Currently, due to financial constraints, the shop cannot afford a digital tracking system. Instead, it relies on memory and paper records, with no proper record for sales, profits, or expenses tracking. Due to the absence of proper and consistent expense and revenue tracking, it is difficult to calculate profits and control unnecessary costs. The lack of workforce stability also affects operations, and the shop struggles to attract more customers due to its distant location.

To solve these issues, a manual yet structured data record-keeping system can be introduced. Using simple tables and written logs, the shop can track sales, profits, and expenses effectively. And also monitor employee trends, and analyze customer patterns. This approach will help retain workers, and increasing profitability without requiring a digital system.

3. Problem Statement

- 3.1 To develop a manual record-keeping system for tracking daily sales, expenses, and profits to improve financial management.
- 3.2 To analyze employee hiring and retention patterns to identify reasons for high turnover and suggest strategies for improving worker stability.
- 3.3 To study customer visits and purchasing trends to develop marketing strategies that attract more students despite shop's distant location from the college.

4. Background of the Problem

The shop operates without any form of record-keeping, relying solely on memory to track sales, expenses, and profits. This lack of structured data makes it difficult to measure profitability, manage costs, or make informed business decisions. Additionally, the shop faces workforce instability, as employees leave frequently, disrupting daily operations. Another key issue is its location, which is 1-2 km away from the college, limiting students visits and affecting sales.

The problem arises due to financial constraints, preventing the shop from adopting digital tracking or relocating to a better area. Without proper data, decision-making is based on guesswork, leading to inefficiencies.

Key Factors Related to the Problem:

- **Major Cause:** Lack of structured data and financial limitations.
- **Internal Problems:** No proper financial tracking, workforce instability, and difficulty in monitoring business performance.
- **External Problems:** The shop's location limits access to college students, reducing potential sales.

A structured manual record-keeping system can help track finances, analyze worker retention, and improve marketing strategies to attract more customers, ultimately increasing profitability.

5. Problem Solving Approach

To address the shop's challenges, structured manual record-keeping and analytical approaches will be used. These techniques are inspired by proven business problem business-solving methods, ensuring better decision-making without requiring a digital system.

1. Manual Financial Tracking

Since the shop does not maintain any records, a simple yet structured logbook will be introduced to track daily sales, expenses, and profits. By consistently recording transactions, the business owner can calculate profits accurately, control unnecessary costs, and set realistic financial goals.

2. Employee Retention Analysis

One of the major challenges faced by the shop is high employee instability. The employee's leave without prior notice leading to disruption in production capacities. To prevent this issue, a particular amount will be reduced from the employee's salary for 3 months as security deposit, which will be returned to him before leaving with a prior notice. This will prevent the disruption in production capabilities and reducing losses.

3. Customer Demand Analysis

The shop relies on memory to track customer demand, which may lead to missed sales opportunities. A simple record of daily sales trends will be maintained to analyze peak hours, best-selling products, and seasonal demand. This will help optimize stock levels, pricing strategies, and menu offerings based on actual customer preferences.

4. Location-based Marketing Insights

The shop is located 1-2 km away from the college or exam center, making it difficult to attract optimal student customers. To overcome this challenge, a temporary pushcart or *thela-gaadi* can be used to make the snacks available far from the shop during recess hours in the college. This will help in improving profitability with minimum investment until they can afford to rent a shop nearby.

5. Cost-Benefit Analysis for Expansion

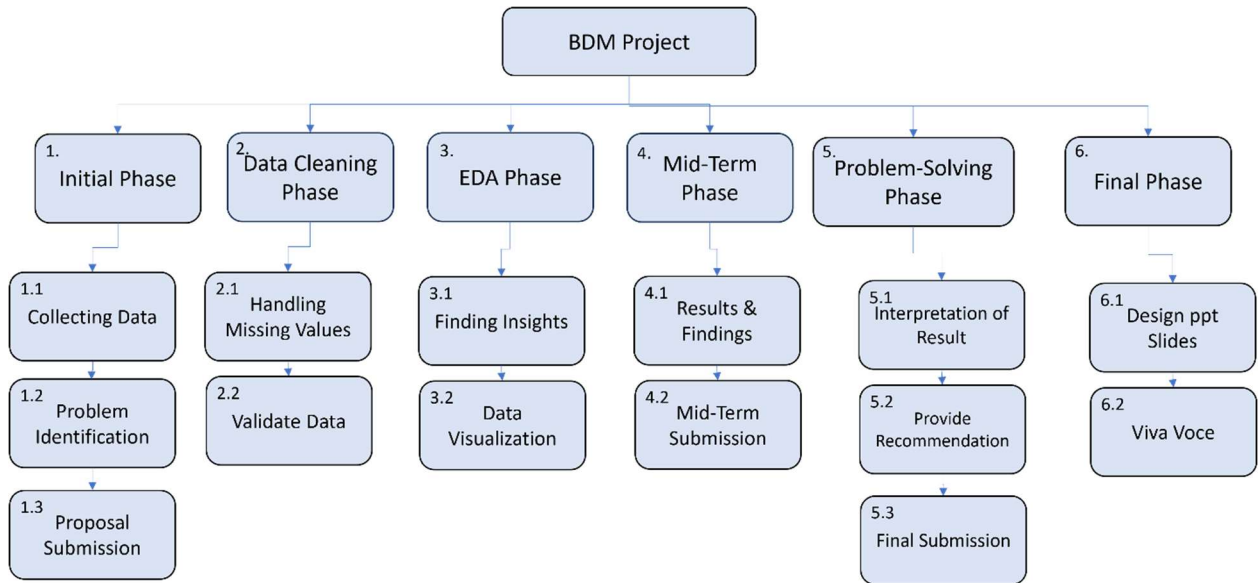
Since the business owner wants to relocate closer to the college but faces financial constraints, a cost-benefit analysis will be conducted using financial records. This will help determine whether expansion is feasible and what strategies can be used to generate the necessary funds, such as increasing high-margin items, making snacks available near college area using pushcart, or reducing unnecessary expenses.

Conclusion

By implementing these structured approaches, the shop can make informed decisions, reduce financial risks, and improve overall efficiency. These methods provide a strong foundation for long-term growth requiring a digital tracking system.

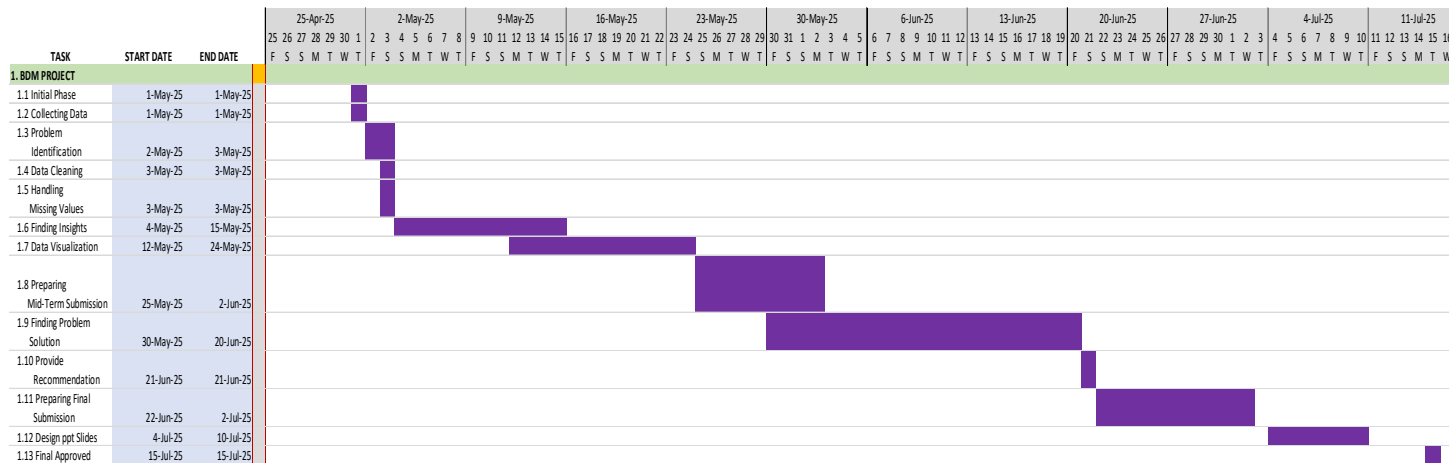
6. Expected Timeline

6.1 Work Breakdown Structure



6.2 Gantt-Chart

PROJECT START DATE: 25 April 2025



7. Expected Outcome

By introducing a simple manual record-keeping system, the business will have a clear understanding of its daily sales, expenses, and profits. This will help in tracking financial growth and controlling unnecessary costs. With better records, the owner can make informed decisions instead of relying on memory.

The system will also help in analyzing customer preferences, showing which food items sell the most. This can guide the owner in planning stock purchases wisely, avoiding unnecessary expenses on less popular items.

Additionally, maintaining data on labor will help in identifying why workers leave frequently. This insight can help in improving working conditions or hiring strategies to retain skilled employees.

Although the shop is not in the ideal location, understanding customer trends can help in planning promotions or discounts to attract more students. Overall, this solution will support better financial management, labor stability, and business growth.